Astronomy & Astrophysics manuscript no. main
July 31, 2026

The supernova history of Cygnus OB2 from *Gaia* DR3

A probabilistic ledger of massive-star deaths from a homogeneous census

V. Voitsekhovskyi[1] and (co-authors to be confirmed)[1]

Affiliation to be completed e-mail: `jstvdk@gmail.com`

Received —; accepted —

ABSTRACT

Context. Cygnus OB2 is one of the most massive young stellar associations in the Galaxy, and its past supernovae shape everything around it: the mechanical energy of the superbubble, the compact-remnant population, the radioactive ^{26}Al budget, and the viability of every interpretation of the region's high-energy emission that invokes a recent explosion. Yet its supernova history has never been measured from a homogeneous census — existing numbers are population-synthesis estimates built on heterogeneous literature inputs.
Aims. We measure the supernova history of Cygnus OB2 as a population property — how many core-collapse supernovae it has produced, their timeline, and the time since the most recent — independently of any single interpretation of the region's high-energy emission.
Methods. From *Gaia* DR3 astrometry and photometry joined to 2MASS and to a literature spectroscopic census, we build a probabilistic member list and its kinematic substructure, fit per-star extinctions against a spectroscopically anchored spatial prior, derive subgroup ages and per-star masses from two isochrone families, and normalize the initial mass function by a response-aware Poisson fit to completeness-corrected star counts between 2 and 8 $M_\odot$. Supernovae are then counted by stellar-evolution turnoff in a Monte Carlo over 2×10^6 population realizations per branch. Discrete modelling choices — isochrone family, extinction law, IMF slope, star-formation duration, explodability — are carried as branches and never averaged. Every marker comparison was pre-registered before it was computed.
Results. Cygnus OB2 resolves into three kinematic subgroups of 4.00, 4.09 and 2.52 Myr: two older and one younger. It has produced $N_{\rm SN} = 8.43$ supernovae on the baseline branch (median 8, 68% [5, 11]), with a carried range of 5.63–28.7 across 36 branches, the first 1.30 Myr ago and the most recent probably within the last 100 kyr ($P = 0.552$). The entire budget lies above 30 $M_\odot$: for any black-hole threshold at or below that mass the ledger returns exactly zero on every branch, so the result is conditional on whether very massive stars explode at all. PSR J2032+4127, whose companion is a star in our own census, settles that branch — a neutron star requires a successful explosion — and its age and kinematics agree with a ledger built without looking at it. As one application of the ledger, the probability that Cygnus OB2 supplied a supernova of the age, progenitor type and location that supernova-driven models of the Cygnus γ-ray cocoon require is $P = 0.323$–0.735 (median 0.533), supported on 18 of 18 branches at $\alpha = 2.0$ and on none at $\alpha = 2.3$.
Conclusions. Cygnus OB2 is an active supernova factory whose entire budget is carried by its most massive stars. The supernova-driven cocoon scenario is supported under those models' own permissive age allowance ($P = 0.727$–0.854) and unresolved at their preferred ~50 kyr, hinging almost entirely on the high-mass IMF slope — an axis *Gaia* DR4 will not settle, but a second association analysed identically would.

Key words. stars: massive – open clusters and associations: individual: Cygnus OB2 – supernovae: general – gamma rays: ISM – astrometry

1. Introduction

Cygnus OB2 is one of the most massive young stellar associations in the Galaxy — tens of O stars, three Wolf–Rayet stars and a young pulsar (Wright et al. 2015; Berlanas et al. 2020) — and the engine of the Cygnus X region. Much of what is astrophysically interesting around it traces back to a single unmeasured population property: how many of its stars have already died, and when. The supernova count and timeline set the mechanical energy injected into the superbubble, the expected compact-remnant population, the radioactive ^{26}Al budget, and the plausibility of every interpretation of the region's high-energy emission that invokes a recent explosion.

That last dependence has become acute. The γ-ray cocoon around Cygnus OB2, detected from GeV to hundreds of TeV, is the clearest Galactic PeVatron candidate, and the proposed accelerators span the association's collective winds, a supernova exploding into the cavity those winds have carved (Härer et al. 2025; (authors to be completed at submission) 2026b), and — since LHAASO's detection of a variable ultra-high-energy source coincident with Cygnus X-3 — a microquasar ((LHAASO-related authors, to be completed at submission) 2026). The field also hosts high-energy sources in their own right: the γ Cygni supernova remnant and the pulsar PSR J2032+4127 with its wind nebula. These candidates are not mutually exclusive, but they place different demands on the stellar population, and the supernova-driven class places the sharpest one: Härer et al. (2025) require an event of 3–5×10^{51} erg from a stripped-envelope progenitor, inside the cavity, within the last few tens to few hundreds of kiloyears. Cygnus OB2 contains three Wolf–Rayet stars, tens of supergiants and a young pulsar, so supernovae have surely occurred; but "surely at least one" is not the same statement as "one within the last 100 kyr", and to date the stronger statement has been supported by plausibility arguments rather than by a census.

The stellar side of the problem is tractable. If the present-day mass function of an association is measured where it is complete, the initial mass function can be normalized there and extrapolated above the turnoff, and the stars that are missing from the upper main sequence are the ones that have died. This is the deficit method of Fuchs et al. (2006) and Zucker et al. (2022), and *Gaia* DR3 makes it possible to apply it to Cygnus OB2 with membership, distance and extinction all derived from the same homogeneous data set rather than assembled from the literature.

Prior estimates exist and disagree in ways that trace to their inputs rather than to their methods: Knödlseder et al. (2002) and Martin et al. (2010) modelled the Cygnus complex as a whole, and Menchiari et al. (2024) adopt a fixed association mass and predict 7 ± 2.5 supernovae at 3 Myr and 26 ± 5 at 5 Myr — a factor of four spanned by a 2 Myr change in an assumed age. Any new number is only useful if it comes with the machinery to say why it differs from the old ones.

This paper supplies that: a measurement of the supernova history of Cygnus OB2 as a property of its stellar population, made prior to — and independently of — any particular interpretation of the high-energy emission. Section 2 describes the data, Sect. 3 the chain from membership to the supernova ledger, and Sect. 4 the results and the external cross-checks. Only then, in Sect. 5, is the measured ledger confronted with the supernova-driven interpretation of the cocoon, alongside what the answer does and does not settle. Three choices govern everything and are stated here rather than buried: discrete modelling choices are *carried as branches and never averaged*; every comparison against an independent supernova marker was *pre-registered* before it was computed; and results that failed are reported as failed.

2. Data

2.1. Gaia DR3 and 2MASS

Two *Gaia* DR3 queries define the working samples: a narrow box ($l \in [77°, 83°]$, $b \in [-1.°5, +4°]$) for the census, and a wide box ($l \in [72°, 88°]$, $b \in [-5°, +8°]$) for runaway traceback. Parallaxes were selected in $[0.35, 1.1]$ mas and $G < 19$ — deliberately generous, because the selection is refined by clustering rather than by cuts. Parallax zero-points follow Lindegren et al. (2021). Near-infrared photometry comes from the official *Gaia* × 2MASS cross-match.

2.2. Spectroscopic anchors and supernova markers

The brightest O stars have unreliable or missing *Gaia* astrometry, so they cannot be allowed to fall out of the census silently. A merged anchor table carries spectral types and, where available, effective temperatures from Wright et al. (2015), Berlanas et al. (2019, 2020) and the Galactic O-Star Catalog; the spectral reclassification of the association's B0 population by (authors to be completed at submission) (2025) postdates our freeze and should be folded in at the DR4 rerun; all 2,112 rows of the *Gaia*-based member catalogue are supplemented by these anchors, which are exempt from the astrometric quality cuts and are treated individually.

Independent supernova markers — PSR J2032+4127 from the ATNF catalogue, γ Cygni (G78.2+2.1) from Green's supernova-remnant catalogue, and the INTEGRAL ^{26}Al measurements of the Cygnus region — were frozen at the data-acquisition stage, *before* any ledger existed. Their comparison statistics were fixed in a pre-registration written before the ledger was run.

3. Method

3.1. Membership and substructure

Members are identified by Gaussian-mixture clustering in $(l, b, \mu_{\alpha*}, \mu_\delta)$, with membership probabilities from a Monte Carlo over each star's full astrometric covariance. Parallax is deliberately *excluded* as a clustering feature: the member sample forms a single population at 1.62 kpc with an intrinsic depth of ~45 pc, which DR3 parallaxes cannot resolve, so including it would manufacture structure rather than find it. The number of components was accepted on seed stability rather than on an information criterion; only $k = 3$ is stable across 50 deterministic seeds.

The gate is conjunctive and was applied before any downstream work: recall of Berlanas et al. (2019) spectroscopic members, the yield of three control fields processed identically, the member count, the spatial compactness of the recovered structure, and a no-percolation criterion. All were met. Of 2,112 catalogue rows, 1,392 have $P > 0.5$ and 1,331 carry a subgroup label: Cyg OB2-A (476), Cyg OB2-B (426) and Cyg OB2-C (429).

We do *not* confirm the two-distance structure of Berlanas et al. (2019) within the member sample, and we state the limit of that non-confirmation precisely: parallax was a selection feature, so a foreground group would have been removed before the test ran. The correct claim is that there is no evidence of two distances *among the members we select*, not that the foreground group does not exist. A parallax-blind membership test is the only way to settle it and is deferred.

3.2. Extinction and ages

Differential extinction is the Cygnus problem, and a single regional A_V is inadmissible. Anchors receive A_V from fixed-$T_{\rm eff}$ intrinsic colours, which avoids the temperature–extinction degeneracy that broadband photometry suffers for hot stars; all other members are fitted against reddened synthetic photometry from the same isochrone families used for the ages, with an anchor-informed spatial prior. That prior's mean is a simple-kriging estimate using a variogram fitted to the anchors themselves, and its width is the unconditional variogram sigma — one internally consistent object, where an earlier version widened the width for distant anchors while still centring on them at full weight.

Subgroup ages come from the de-reddened colour–magnitude diagrams with an unresolved-binary population included in the likelihood rather than clipped away. These upper-main-sequence ages (3.98, 3.55, 2.51 Myr) are *the prior*, not the adopted values: the ages used downstream are the counts-based ages of Sect. 3.3.

3.3. IMF normalization

The normalization k is fitted per subgroup by a Poisson likelihood over mass bins between 2 and 8 $M_\odot$, against a forward model that convolves the assumed IMF with a completeness response measured by injecting synthetic stars into the real field and recovering them through the actual selection. The response, not a scalar completeness, is the correct object: the mass estimator has 22–25% scatter, so it scatters stars across bin edges and across the window boundaries in both directions.

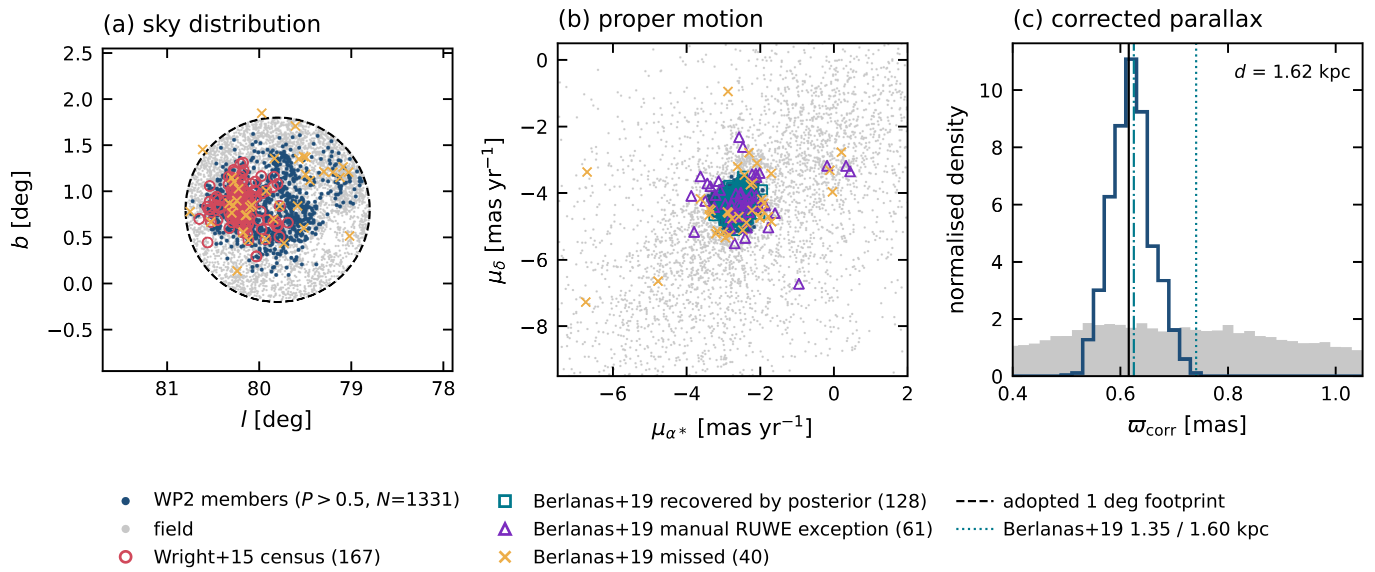

Fig. 1. Membership and substructure. Sky distribution and proper-motion plane of the 1,392 members with $P > 0.5$, coloured by kinematic subgroup, with the Berlanas et al. (2019) and Wright et al. (2015) literature samples overlaid. Parallax is deliberately excluded as a clustering feature (Sect. 3.1).

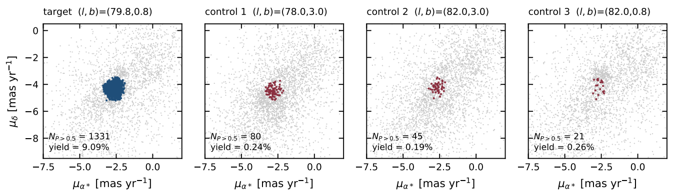

Fig. 2. The false-positive rate, measured rather than assumed. Three control fields at matched Galactic latitude were processed through the identical selection, clustering and membership model; their yield is 3.7% of the association field's.

The truth-side age is marginalized jointly with k. The forward response is a mixture over nine age nodes drawn from the CMD posterior, reweighted by the Poisson likelihood of the observed counts with k integrated out analytically. This produces a *counts-based* age — the age at which the injected population reproduces the observed mass function — which is what the ledger consumes. On the baseline branch these are 4.00, 4.09 and 2.52 Myr, with normalizations $k = 1{,}734$, 1,647 and 1,894.

Only Cyg OB2-B moves materially between the two age definitions, by +0.54 Myr, and that shift is worth a factor 1.92 in B's supernova count — which is why both are reported and labelled. B's counts-based posterior concentrates 53% of its weight on the topmost of its nine nodes, so its age is bounded from below by the data and from above by the prior grid; its supernova contribution is correspondingly a lower bound (Sect. 5.4).

3.4. Census closure and runaways

The closure test compares the number of living stars above 8 $M_\odot$ predicted by the normalization against the number observed. It is an *out-of-sample* test: k is fitted from 2–8 $M_\odot$ counts and the $> 8\,M_\odot$ census never enters that likelihood. Both sides are computed with the same forward response, integrated with no lower cut-off, because the observed side counts every member's $P(M > 8\,M_\odot)$ whatever its true mass.

Runaways are recovered from the wide box by tracing proper motions backwards over the subgroup age windows, with the association's systemic motion subtracted — Cygnus OB2's bulk motion is larger than a typical ejection signature, so absolute proper motions measure drift rather than ejection. The false-positive rate is measured per separation bin from control fields, in the same spirit as recent Gaia runaway searches in other associations ((authors to be completed at submission) 2026a).

3.5. The supernova ledger

For each branch and each of 2×10^6 iterations we draw a paired (k, age) sample from the joint posterior — paired, because the two are correlated through the fit — draw a star-formation duration, sample a discrete stellar population from the IMF conditioned on the calibration-window counts, and ask which stars have passed the turnoff. Supernovae are then those deaths that satisfy the explodability branch.

Runaways do *not* enter $N_{\rm SN}$. They are living stars below the turnoff, so they change neither k, which is fitted below 8 $M_\odot$, nor the turnoff mass. What they bound is location, not number (Sect. 4.5).

3.6. Branches, and what is never averaged

Isochrone family {PARSEC, MIST} × extinction law $R_V \in \{3.0, 3.1, 3.5\}$ × IMF slope $\alpha \in \{2.0, 2.3, 2.6\}$ × star-formation duration $\delta \in \{0, 1, 2\}$ Myr × explodability {all explode, islands of implosion}. The baseline branch is PARSEC, $R_V = 3.1$, $\alpha = 2.3$, $\delta = 0$, all explode. 40 of 54 mass-function fits pass the residual gate; no branch is ever dropped for failing it, and the failures are reported by their concentration in the $R_V = 3.5$ and $\alpha = 2.6$ corners.

The *headline* branch set excludes $\alpha = 2.6$. The criterion is the calibration-window χ^2, an internal gate statistic fixed long before the ledger existed, on which $\alpha = 2.6$ is the best fit in 1 of 18 cells and has the worst median ($\chi^2 = 10.80$ against 6.86 for $\alpha = 2.3$). The census closure agrees independently

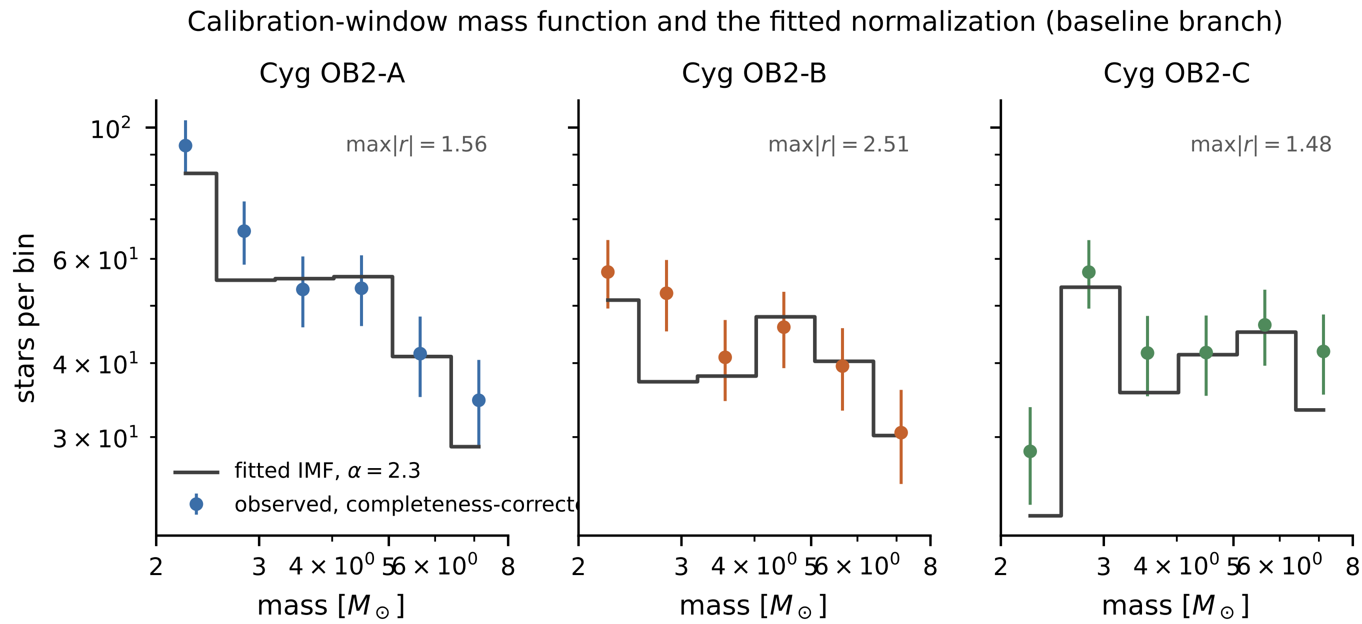

Fig. 3. The calibration-window mass function per subgroup on the baseline branch, with the fitted normalization overlaid. The fit is a Poisson likelihood against a forward model that convolves the assumed IMF with the measured completeness response, not against a completeness-divided histogram. Worst Pearson residuals are annotated.

but was deliberately *not* used to make this choice, so that it remains available as an out-of-sample validation of the extrapolation (Sect. 4.4). $\alpha = 2.6$ is reported throughout with its own numbers ($N_{\rm SN} = 1.93$–4.24, $P_{\rm verdict} = 0.142$–0.250) and is never deleted.

4. Results

4.1. A revised star-formation history

Cygnus OB2 is not coeval. Its three kinematic subgroups have counts-based ages of 4.00, 4.09 and 2.52 Myr — **two older subgroups and one younger**, with a total spread of 1.57 Myr. This differs from the structure the same pipeline returned before the extinction prior was corrected, which read as one older subgroup and two younger; the change followed a pre-registered fix whose four predictions were confirmed in advance, with Cyg OB2-B moving by +0.73 Myr and A and C by 0.000 Myr.

The supernova timeline follows directly. A and B supply the entire budget — their turnoffs stand at 58 and 56 $M_\odot$ — while Cyg OB2-C's sits at 279 $M_\odot$, still above the IMF's upper limit, so it has lost no stars at all.

The wider envelope across both retained age indicators is 2.25–5.67 Myr, and that — not the spread between the three central values — is the age uncertainty everything downstream inherits (Fig. 5).

4.2. The ledger

On the baseline branch Cygnus OB2 has produced

$$N_{\rm SN} = 8.43 \quad (\text{median } 8,\ 68\%\ [5, 11]), \tag{1}$$

comprising 4.17 from Cyg OB2-A, 4.26 from Cyg OB2-B and exactly 0.00 from Cyg OB2-C. The probability that at least one supernova has occurred is 0.9997; the probability that the most recent one falls within the last 100 kyr is 0.552, and the median time since it is 86 kyr. The first explosion was 1.30 Myr ago and the rate has been roughly flat at 8.0 $\rm Myr^{-1}$ since (Fig. 4).

The number is not 8.43. Across the 36 headline branches the total spans 5.63–28.7, a factor of 5.1; including $\alpha = 2.6$ it reaches down to 1.93, a factor of 14.9. The branch spread *is* the uncertainty; the Poisson interval on any one branch is the smaller part of it. The same is true of the recent-supernova probability, which spans 0.411–0.889 across the headline branches. We quote the baseline value with its carried range beside it and never an average across branches, which for a set that carries no probability weights would not be a posterior summary.

4.3. The whole budget lies above 30 $M_\odot$

Every star that has died in Cygnus OB2 is very massive. On the baseline branch the lowest progenitor that exploded is 52 $M_\odot$, and the lowest turnoff any star was compared against is 45.4 $M_\odot$; across the whole grid, on the widest star-formation-window branches, those fall to 33.9 $M_\odot$ and 33.9 $M_\odot$, where up to 19% of the supernovae come from progenitors below 52 $M_\odot$. Consequently, **for any black-hole threshold at or below 30 $M_\odot$ the ledger returns exactly zero supernovae, on every branch and in every iteration**; at a 40 $M_\odot$ threshold it is exactly zero on every coeval branch and at most 1.8% of $N_{\rm SN}$ elsewhere.

This is not a marginal systematic but the shape of the result. The interleaved pattern of explodable and non-explodable ZAMS masses that Sukhbold et al. (2016) and Ertl et al. (2016) find between roughly 15 and 25 $M_\odot$ is *irrelevant* here, because nothing in that range has had time to die — and the 30 $M_\odot$ floor lies above the whole of that structure. The supernova history of Cygnus OB2 is instead entirely conditional on whether stars well above 30 $M_\odot$ explode at all, which these data cannot address — but which an independent marker can (Sect. 4.5).

The same statement generalizes: any association of a few Myr has a turnoff far above the island structure, so its supernova

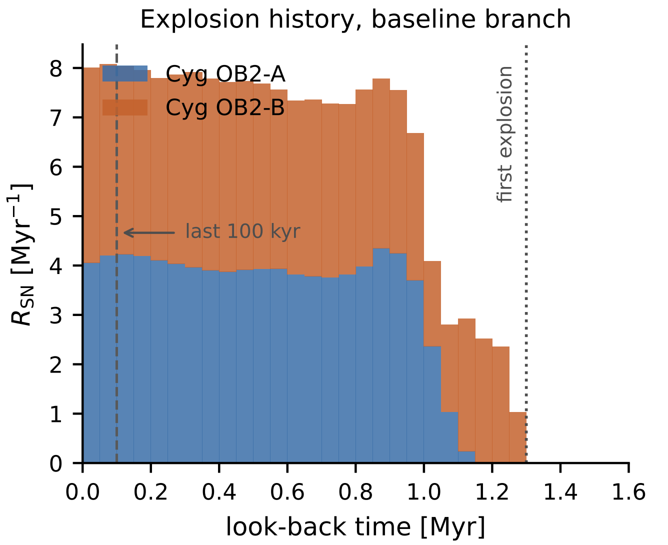

Fig. 4. The explosion history $R_{SN}(t)$ on the baseline branch, stacked by subgroup. Cyg OB2-C contributes nothing: its turnoff has not yet fallen below the IMF's upper limit. The rate is roughly flat since the first explosion 1.30 Myr ago, which is why $P(t_{last} < 100\,\text{kyr}) = 0.552$.

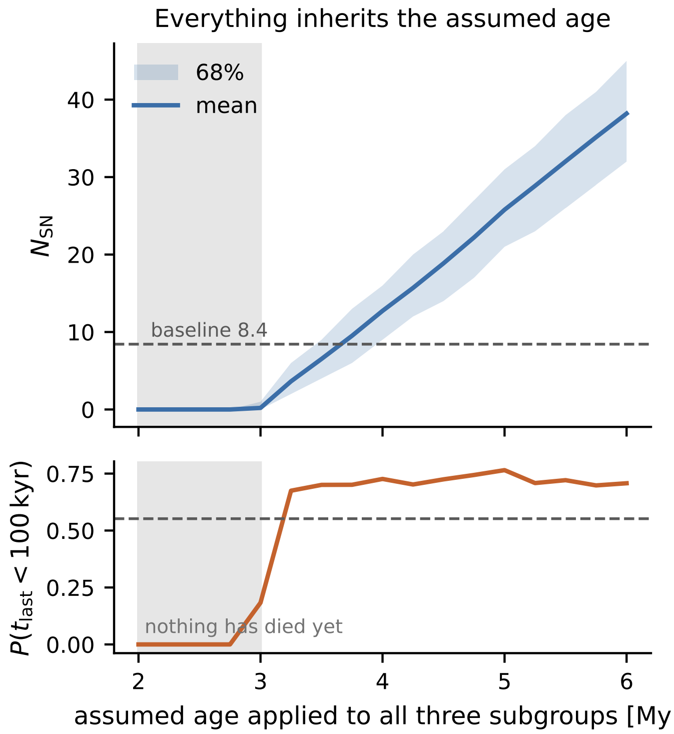

Fig. 5. The mandatory honesty plot: N_{SN} and $P(t_{last} < 100\,\text{kyr})$ as functions of a common assumed age applied to all three subgroups. Below 3.00 Myr the turnoff has not crossed the IMF ceiling and nothing has died; above it the count rises by roughly 12 supernovae per Myr of assumed age, reaching 38.2 at 6 Myr. Everything this paper reports inherits the ages of Sect. 3.3, and this is how that dependence is disclosed rather than absorbed.

budget is a measurement of very-massive-star explodability and not of the canonical core-collapse regime.

4.4. The census closes at Salpeter

The strongest objection to any deficit method is that the IMF slope is constrained where the counts are complete and used where they are not. Here that objection has an answer, because the closure test never entered the fit.

At $\alpha = 2.3$ the predicted and observed counts of living stars above $8\,M_\odot$ agree to 6.7% (grid median 1.067), and the slope at which the census closes exactly is $\alpha \approx 2.25$, with 18 of 18 cells inside the carried branch grid. A single power law from 2 to $120\,M_\odot$ is therefore not assumed; it is tested, and it holds at the few-per-cent level. **This is the payoff of not spending the closure test to select** α: had we done so, this answer would have been circular.

The residual is not uniform. Cyg OB2-A closes at 0.865, B at 1.068 and C at 1.360. C's excess is the fourth independent appearance of a "Cyg OB2-C is different" signal, alongside its shallower closing slope, its behaviour in the multiplicity test and its preference for a shallower slope on both internal and out-of-sample evidence. We report it as an open science question rather than absorbing it; the leading candidate is distance contamination, which would inflate masses and therefore closure ratios in exactly C's direction, and which no alternative we tested addresses.

The living census totals 380.6 stars above $8\,M_\odot$: 295.4 in the member channel, 27 spectroscopic anchors inside the footprint but outside the member sample, and 58.2 runaways — so 322.4 were retained and at most 14.6% of the ledger's supernovae were not in situ. The runaway search recovers 119 raw candidates, 54.9 after the measured false-positive correction, and it recovers the canonical ejected star BD+43°3654 at unit probability with $38.8\,\text{km}\,\text{s}^{-1}$ and a 1.36 Myr look-back time against literature values of $\sim 40\,\text{km}\,\text{s}^{-1}$ and 1.6 Myr — from an estimator that knows nothing about the published answer.

4.5. External cross-checks

Every comparison in this section was pre-registered, with its statistic and its pass condition fixed, before the ledger existed. All five predictions passed.

PSR J2032+4127 — the marker that settles a branch. The pulsar's Be-star companion MT91 213 is *our own census star* (*Gaia* DR3 2067835682818358400), a B0V of $17\,M_\odot$ lying 0°.115 from the Cyg OB2-A centroid and already counted in the living census. A neutron star requires a *successful* explosion, and this is therefore direct evidence against the branch on which the budget is identically zero: the ledger gives $P(\geq 1\,\text{SN}) = 0.9997$ if massive stars explode against exactly 0 on the islands branch. The agreement extends further than existence. The pulsar's characteristic age, widened for the line-of-sight acceleration its decades-long eccentric orbit imposes, gives 151–401 kyr, against a ledger probability of 0.960 that the last supernova falls inside that window; and its peculiar transverse velocity of $27.6\,\text{km}\,\text{s}^{-1}$ is as low as a still-bound binary requires.

Three readings of the pulsar remain degenerate and we do not pretend otherwise: high-mass explodability, an older population than we fit, or binary stripping of a lower-mass progenitor. *All three give a non-zero budget*, which is what the argument needs.

γ Cygni — allowed, unsettled. The ledger gives $P = 7.7\%$ for a supernova within the remnant's $\sim$7 kyr age, so a physical association is allowed but not required. Our association distance of 1.62 kpc overlaps the remnant's 1.7–2.6 kpc range only at its low extreme. We leave this unsettled by design.

Absent remnants — weak evidence. Even at a generous 100 kyr visibility lifetime and with no cavity, the ledger predicts only 0.80 visible remnants. The absence of a catalogued remnant is therefore consistent with the cavity argument but does not test it.

^{26}Al — a consistency band. ^{26}Al traces both supernova ejecta and Wolf–Rayet winds, so it constrains the combination and cannot be inverted for a supernova count. It is reported as a band and never used as a fit. Run forward rather than backward — and, unlike everything else in this section, post-hoc rather than frozen at the outset (Sect. 5.7) — the ledger's supernovae alone supply 10–42% of the measured complex-wide 1809 keV flux (Martin et al. 2009). That is a lower bound on the measurement, since the winds add to it: supernovae are a sub-dominant but not negligible contributor.

5. Discussion

5.1. Application: the supernova-driven cocoon scenario

The ledger was measured without reference to any high-energy interpretation of the region; it can now be confronted with the interpretation that demands the most of it. Evaluating the requirement of Härer et al. (2025) — an event of the right age, progenitor type and location (Sect. 1) — over the 36 headline branches gives

$$P_{\rm verdict} = 0.323-0.735 \quad (\text{median } 0.533), \tag{2}$$

decomposing into an age term $C_1 = 0.379$–0.861, a progenitor-type term $C_3 = 1.000$ and an in-situ term $C_4 = 0.854$. The explosion-energy condition is deliberately *not* multiplied in: this analysis measures progenitor masses, not explosion energies, and refuses to invent a probability for them. $P_{\rm verdict}$ is therefore the probability that a supernova of the right age, type and location was available, *given* that such an event can reach the required energy — and the two requirements point the same way, since an energetic stripped-envelope progenitor is exactly what the ledger contains.

$C_3 = 1.000$ exactly. Every supernova came from a progenitor above 33.9 $M_\odot$ on every branch, and above 52 $M_\odot$ on the coeval ones, so all of them should have been stripped-envelope type Ib/c events — the channel the PeV-bubble model requires. That is not a coincidence but a restatement of Sect. 4.3.

The verdict hinges on one axis. The spread in $P_{\rm verdict}$ is 0.264 across α, against 0.065 across R_V, 0.045 across isochrone family and 0.043 across star-formation duration. The split is exact: every one of the 18 $\alpha = 2.0$ branches clears 0.5 (0.593–0.735) and none of the 18 $\alpha = 2.3$ branches does (0.323–0.474), with the boundary falling precisely between them (Fig. 6).

The framing rule — fixed at planning time and operationalized before the number was known — returns INCONCLUSIVE, and hence this article rather than a Letter.

Under the model's own permissive window. Härer et al. (2025) themselves allow an event "within the last few hundred kyr". Against that reading every branch clears 0.5, P = 0.727–0.854. The scenario is therefore *supported* against its authors' permissive statement and *unresolved* against their preferred ∼50 kyr. Both windows were pre-registered before either was computed.

5.2. $P \approx 0.5$ is a measurement, not a coin flip

A coin returns 0.5 to every question. This analysis does not: it returns 0.055 for a 7 kyr window, 0.552 for 100 kyr and 0.854 for the permissive one. The relevant comparison is against ignorance. A last supernova distributed uniformly over the association's 1.30 Myr active period would fall inside 100 kyr with probability 0.077; we measure 0.552, a factor of 7 higher, because the rate is measured rather than assumed. The number is near one half because the answer genuinely sits near the boundary, not because the analysis is uninformative.

5.3. Binary mass transfer, bounded

Our count is a single-star turnoff calculation, and most massive stars have companions — Sana et al. (2012) find a close-binary fraction near 0.7 for 15–60 $M_\odot$, and about two-thirds of Cygnus OB2's own O and early-B stars are resolved or spectroscopic multiples. Two independent lines bound what that omission costs.

Empirically, the BPASS (Stanway & Eldridge 2018) calculation of Härer et al. (2025) — *with* binaries, for a population of the same mass and IMF slope — gives a core-collapse rate at 4 Myr that is 0.72–0.73 times ours once the difference in IMF integration ranges is corrected. The binary-inclusive rate sits *below* our single-star one, not above. Theoretically, Zapartas et al. (2017) find that binarity raises the integrated number of core-collapse supernovae by 14^{+15}_{-14}%, but attribute the increase to *late* events, 50–200 Myr after birth, from 4–8 $M_\odot$ binaries, and find the early delay-time distributions "remarkably similar", diverging only around 20 Myr. Cygnus OB2 is 4 Myr old and its first supernova was 1.30 Myr ago, so the entire ledger sits inside the regime where the two distributions agree, and the channel producing the +14% cannot have operated at all.

Nor does the living binary fraction translate directly. That 30^{+10}_{-15}% of massive main-sequence stars are binary products (de Mink et al. 2014) is a statement about survivors: accretors and mergers are rejuvenated, so a star pushed above the present turnoff by mass transfer has had its clock partly reset and has typically *not* yet exploded.

Adopting a deliberately conservative ±30% bracket — the top of the Zapartas et al. (2017) enhancement applied in full despite its channel being inapplicable — and running it through the ledger engine moves the baseline from 5.90 to 10.96 supernovae, a span of 5.1 against a headline branch span of 23.1. **The systematic we do not model is about 5 times smaller than the model-branch spread we do report.** The upward arm helps the PeV-bubble scenario, so the verdict is conservative with respect to it.

What binaries *do* change is the supernova type, and there the effect is already in the answer: $C_3 = 1.000$. That does not smuggle binaries into the result, because at 34–58 $M_\odot$ single-star Wolf–Rayet winds strip envelopes too, and Cygnus OB2 contains three known WR stars.

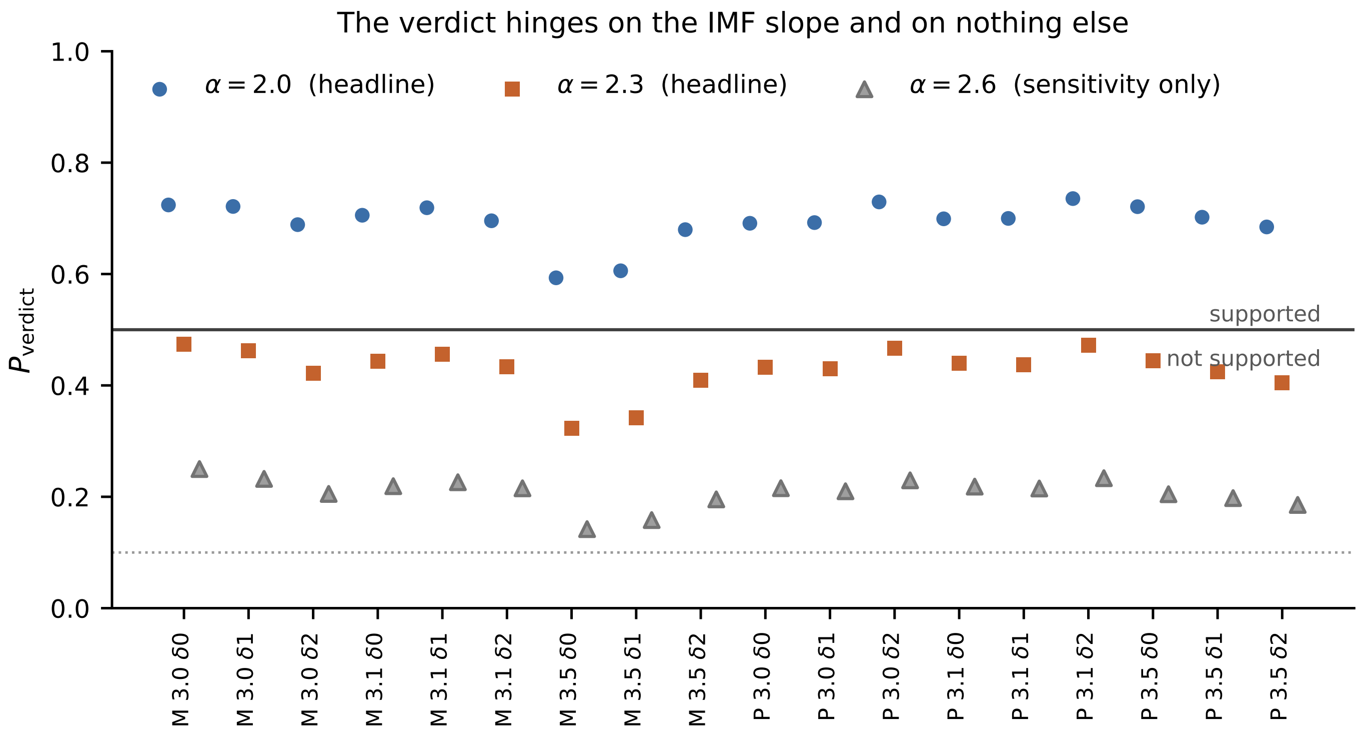

Fig. 6. $P_{\rm verdict}$ against every branch. The headline set (α = 2.0 and 2.3) is coloured and the sensitivity-only α = 2.6 branches are grey; the solid line is the 0.5 support boundary and the dotted line the 0.1 disfavour boundary of the pre-registered framing rule. Every α = 2.0 branch clears 0.5 and no α = 2.3 branch does, with the boundary falling precisely between the two slopes — which is why the rule returns INCONCLUSIVE and what future work must settle.

5.4. Systematics budget

The absolute extinction scale. At matched sky position, prior-free broadband photometry prefers an A_V about 0.5 mag lower than the spectroscopically calibrated anchors ($p = 4.3 \times 10^{-16}$). The absolute extinction scale, and hence the mass scale, the normalization and $N_{\rm SN}$, rests on the anchor calibration rather than on the photometry alone. This is a systematic on the *scale*, not on the relative structure: it moves all three subgroups together and does not produce the A/B/C ordering. Its sign was checked and is the reassuring one — the anchors read *higher*, the opposite of the failure mode in which a selection-biased anchor set would drag the scale down.

Cyg OB2-B's calibration asymmetry. B contains 4 spectroscopic anchors inside the member sample against 59 for A and 42 for C, and its eighth-nearest anchor lies $0\overset{\circ}{.}374$ away against $0\overset{\circ}{.}089$ and $0\overset{\circ}{.}139$. Its extinction therefore rests on weaker local calibration than its siblings'. This is an observational limitation of the DR3-era anchor set, not a modelling choice: the kriging weights sum to 1.000 for A and 0.992 for C but only 0.772 for B, so the correction is intrinsically B-specific and no per-subgroup decision was made. The same fact surfaces three times — as B's extinction correction, as the last mass-function gate blocker, and as B's age railing against its prior support.

B's age is a one-sided bound. 53% of B's counts-based age posterior sits on the topmost of its nine prior nodes. A posterior that piles up against the edge of its prior grid reports a bound, not a measurement, and the direction is known: older means a lower turnoff means *more* supernovae. Were B's true age the top node (4.26 Myr) its supernova count would rise by 15%. B's contribution to $N_{\rm SN}$ is therefore a lower bound at the precision quoted.

The in-situ fraction is an upper bound. The runaway count is a lower bound twice over — the traceback is two-dimensional, so line-of-sight ejections are missed, and the wide box caps recovery at 44 km s^{-1} over 5 Myr. A higher true runaway fraction lowers C_4 and therefore the verdict. The effect is bounded: C_4 would have to fall from 0.854 to below 0.60 to move any α = 2.0 branch under 0.5. The direction is against us and we state it.

Association mass, and which one. Three nested integrals of the same normalization can all be called "the association mass", and they differ by a factor of 1.67. We quote the multiplicity-adjusted stellar mass over 0.08–120 $M_\odot$, 2.92×10^4 $M_\odot$. The like-for-like partner of Wright et al.'s 2015 1.65×10^4 $M_\odot$ — primary stars over the full range — is 2.42×10^4 $M_\odot$, a factor 1.47. Integrating only 0.5–120 $M_\odot$ instead gives 1.75×10^4 $M_\odot$, which coincides with the literature value but compares a different quantity. The comparison is an agreement at the factor level, which is the resolution IMF-extrapolated cluster masses admit.

5.5. Pre-registration as part of the result

Every marker comparison in Sect. 4.5 was pre-registered, with its statistic and pass condition fixed, before the ledger it tests existed; the markers themselves were frozen at data acquisition, before the normalization, the census and the ledger were written. The question "did you tune the analysis to the markers?" therefore has a checkable answer rather than a reassurance. The same discipline produced results we would rather not have: predictions recorded as failed and not reinterpreted, and two intermediate results withdrawn after defects were found by comparison against externally known values (Appendix C). We report this not as virtue but because the verdict's credibility rests on it: an analysis that can be tuned to a ~50 kyr window is not evidence about a ~50 kyr window.

5.6. What Gaia DR4 will and will not fix

DR4 improves three things that matter here: radial velocities for bright OB stars make the traceback three-dimensional, so runaways stop being a lower bound and C_4 tightens; better astrometry and photometry sharpen the subgroup ages, including Cyg OB2-C's family-dependent 2.52-versus-3.16 Myr ambiguity and the 0–7 supernova range that follows from it; and deeper membership assigns subgroups to the currently unlabelled massive members and orphan anchors. Gaia XP spectra address Cyg OB2-B's anchor famine specifically.

DR4 will not settle the dominant term. The verdict hinges on the high-mass IMF slope, with a spread of 0.264 against 0.045 for everything astrometric. Distinguishing $\alpha = 2.0$ from 2.3 needs a larger or deeper massive-star census, or a second association analysed identically — not better astrometry on this one. We state the asymmetry because it is what makes the forecast usable.

5.7. A forward prediction for COSI

Since DR4 will not settle α, we make the ledger falsifiable somewhere else. *Unlike the cross-checks of Sect. 4.5, this comparison was not frozen before the analysis began: it is a post-hoc addition, chosen after the ledger existed and pre-registered before it was scored.* We flag it because the weight the frozen markers carry depends on that distinction.

^{60}Fe is the supernova-specific tracer — Wolf–Rayet winds make none of it, so unlike ^{26}Al it counts explosions rather than a combination. Carrying published per-supernova yields as a branch fixed before any flux was computed, and accumulating each event's ejecta at its own progenitor mass with decay from its own epoch, the headline branches predict 3.0–$14.3 \times 10^{-3}\,M_\odot$ of ^{60}Fe surviving now, beside 0.92–$3.7 \times 10^{-3}\,M_\odot$ of supernova ^{26}Al. The association is nowhere near steady state — only a fraction 0.16–0.31 of the saturated ^{60}Fe mass has accumulated, because the first supernova is more recent than the isotope's mean lifetime — so the usual rate-times-lifetime estimate would overstate the signal several-fold.

At 1.62 kpc that gives a median 1173 keV line flux of $3.6 \times 10^{-6}\,\mathrm{ph\,cm^{-2}\,s^{-1}}$, against COSI's expected 3σ narrow-line sensitivity of 3.0×10^{-6} in a two-year survey (Tomsick & COSI Collaboration 2023). The split is clean: 18 of 18 branches with $\alpha = 2.0$ clear that sensitivity and 0 of 18 with $\alpha = 2.3$ do, a factor 2.68 between the medians. **COSI would discriminate the one axis this paper cannot decide and better astrometry will not fix.**

The forecast is ^{60}Fe-rich by Galactic standards — the predicted ^{60}Fe/^{26}Al flux ratio is 0.76–1.00, against 0.184 ± 0.042 measured for the Galaxy (Wang et al. 2020) — but the two are not the same quantity, and the difference is expected rather than anomalous: our ^{26}Al denominator counts supernova ejecta only, where the Galactic ratio's includes wind ^{26}Al, and every progenitor here lies above $30\,M_\odot$, where the ^{60}Fe yield peaks.

Two things bound that claim, and we give them equal weight. The yield model matters more than our census does: the spread between published yield arms is a factor 30 in the predicted flux, against 4.8 across the entire headline branch set (Limongi & Chieffi 2006), so a measurement would constrain the two jointly rather than α alone. And the current standard compilation predicts nothing at all — Limongi & Chieffi (2018)'s recommended scenario collapses every star above $25\,M_\odot$ straight to a black hole, and 0 of the 315 million supernovae we sample fall below that threshold, so on that prescription this ledger's ^{60}Fe yield is identically zero (Falla et al. 2025). A non-detection would then favour precisely the explodability that PSR J2032+4127 disfavours, and that tension would itself be the result. Existing data are already near the boundary: our richest branch predicts a combined ^{60}Fe flux within 5% of the INTEGRAL/SPI upper limit for the Cygnus region, $1.6 \times 10^{-5}\,\mathrm{ph\,cm^{-2}\ s^{-1}}$ (Martin et al. 2009).

5.8. Deferred

Three questions are named here and left for later work rather than held against this paper: a parallax-blind membership test, which is the only way to break the circularity in our non-confirmation of the two-distance structure; a distance-contamination test for Cyg OB2-C, the leading explanation of its closure excess; and the resolution of C's 0-versus-7 supernova family ambiguity, which DR4 ages should settle.

6. Conclusions

1. Cygnus OB2 resolves into three kinematic subgroups of 4.00, 4.09 and 2.52 Myr — two older and one younger — at 1.62 kpc, with no evidence for two distinct distances among its members.
2. It has produced $N_{\mathrm{SN}} = 8.43$ supernovae on the baseline branch (median 8, 68% [5, 11]), with a carried range of 5.63–28.7 across 36 branches. The first was 1.30 Myr ago; the most recent falls within the last 100 kyr with probability 0.552.
3. The entire budget lies above $30\,M_\odot$. For any black-hole threshold at or below that mass the ledger returns exactly zero on every branch, so the result is a measurement of very-massive-star explodability, and the 15–25 $M_\odot$ island structure is irrelevant at this age.
4. PSR J2032+4127 — whose companion is a star in our own census — settles that branch: a neutron star requires a successful explosion. Its age and kinematics agree with a ledger built without looking at it.
5. The census of stars above $8\,M_\odot$ closes at $\alpha \approx 2.25$ and to 6.7% at Salpeter, an out-of-sample validation of the IMF extrapolation that was deliberately not spent on selecting α.
6. Applied to the supernova-driven interpretation of the Cygnus cocoon, $P = 0.323$–0.735 (median 0.533): supported on all $\alpha = 2.0$ branches, on no $\alpha = 2.3$ branch, and supported on every branch (0.727–0.854) under the model's own permissive age allowance. The verdict is a measurement, not a coin flip: against an ignorance baseline of 0.077 it is a factor 7.
7. Unmodelled binary mass transfer is bounded at ±30% of N_{SN}, about 5 times smaller than the branch spread already reported.

Acknowledgements. This work has made use of data from the European Space Agency (ESA) mission *Gaia* (https://www.cosmos.esa.int/gaia), processed by the *Gaia* Data Processing and Analysis Consortium (DPAC, https://www.cosmos.esa.int/web/gaia/dpac/consortium). Funding for the DPAC has been provided by national institutions, in particular the institutions participating in the *Gaia* Multilateral Agreement. This publication makes use of data products from the Two Micron All Sky Survey, and of NASA's Astrophysics Data System.

References

(authors to be completed at submission). 2025, arXiv e-prints [2510.15540]
(authors to be completed at submission). 2026a, arXiv e-prints [2604.11988]
(authors to be completed at submission). 2026b, arXiv e-prints [2606.03881]
Berlanas, S. R., Herrero, A., Comerón, F., et al. 2020, A&A[2008.09917]
Berlanas, S. R., Wright, N. J., Herrero, A., Drew, J. E., & Lennon, D. J. 2019, MNRAS, 484, 1838
de Mink, S. E., Sana, H., Langer, N., Izzard, R. G., & Schneider, F. R. N. 2014, ApJ, 782, 7
Ertl, T., Janka, H.-T., Woosley, S. E., Sukhbold, T., & Ugliano, M. 2016, ApJ, 818, 124
Falla, A., Roberti, L., Limongi, M., & Chieffi, A. 2025, ApJ[2508.07705]
Fuchs, B., Breitschwerdt, D., de Avillez, M. A., Dettbarn, C., & Flynn, C. 2006, MNRAS, 373, 993
Härer, L., Vieu, T., Schulze, F., Larkin, C. J. K., & Reville, B. 2025, A&A, 703, A111
Knödlseder, J., Cerviño, M., Le Duigou, J.-M., et al. 2002, A&A, 390, 945
(LHAASO-related authors, to be completed at submission). 2026, arXiv e-prints [2607.07100]
Limongi, M. & Chieffi, A. 2006, ApJ, 647, 483
Limongi, M. & Chieffi, A. 2018, ApJS, 237, 13
Lindegren, L., Bastian, U., Biermann, M., et al. 2021, A&A, 649, A4
Martin, P., Knödlseder, J., Diehl, R., & Meynet, G. 2009, A&A, 506, 703
Martin, P., Knödlseder, J., Meynet, G., & Diehl, R. 2010, A&A, 511, A86
Menchiari, S., Morlino, G., Amato, E., Bucciantini, N., & Beltrán, M. T. 2024, A&A[2402.07784]
Sana, H., de Mink, S. E., de Koter, A., et al. 2012, Science, 337, 444
Stanway, E. R. & Eldridge, J. J. 2018, MNRAS, 479, 75
Sukhbold, T., Ertl, T., Woosley, S. E., Brown, J. M., & Janka, H.-T. 2016, ApJ, 821, 38
Tomsick, J. A. & COSI Collaboration. 2023, in Proc. 38th International Cosmic Ray Conference (ICRC2023), 745
Wang, W., Siegert, T., Dai, Z. G., et al. 2020, ApJ, 889, 169
Wright, N. J., Drew, J. E., & Mohr-Smith, M. 2015, MNRAS, 449, 741
Zapartas, E., de Mink, S. E., Izzard, R. G., et al. 2017, A&A, 601, A29
Zucker, C., Goodman, A. A., Alves, J., et al. 2022, Nature, 601, 334

Appendix A: Branches, gates and their verdicts

Appendix B: Provenance and reproducibility

Every cut, threshold, catalogue version and query is recorded with its date in the project's provenance log, and every gate verdict exists as a machine-readable record carrying SHA-256 digests of its inputs and outputs. Nothing is overwritten: superseded artifacts are retained with version suffixes and stay referenced. The manuscript's numbers are generated directly from those artifacts; no number in this paper was typed by hand.

Appendix C: Withdrawn and failed results

Two intermediate results were withdrawn after defects were found, both by comparison against externally known values rather than by internal checks: a runaway census computed from absolute rather than peculiar proper motions, and a set of closure ratios whose forward integral was truncated at the same threshold its response scatters across. Four pre-registered predictions were recorded as failed and not reinterpreted. Both classes are listed here because a chain that reports only its successes cannot be audited.